Table 1: Saturation parameters by stone.

Stone	τ (min)	$D_{\max}$
Duan (Laokeng)	4.1	0.94
Duan (Majiakeng)	6.8	0.87
She	9.2	0.81

Ink Density Saturation in Duan Inkstones

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Abstract

A minimal document in the NeurIPS 2026 format. The content mirrors `latex-rendering-check.tex`: ink density saturates as a function of grind time, and the stone — not the stick — sets the ceiling.

1 Model

Let $D(t)$ denote ink density after t minutes of grinding. Density follows a saturating exponential,

$$D(t) = D_{\max}\left(1 - e^{-t/\tau}\right), \tag{1}$$

where τ is the stone's time constant and $D_{\max}$ its ceiling. The relationship was established empirically by Tanaka and Sato [2019], and half saturation is reached at ${}^{\ln 2}/{}_{1} \cdot \tau$.

2 Results

Table 1 lists time constants for three stones. The grinding surface, not the ink stick, dominates [Mizuta, 1968].

3 Checklist

Opening this file and pressing preview should produce a single-column PDF with line numbers down the left margin, the NeurIPS footer on page one, a numbered equation, a booktabs table, and resolved citations to Tanaka and Sato [2019] and Chen and Nakamura [2021].

References

Liang Chen and Aoi Nakamura. A survey of provenance methods for She and Duan inkstones. In *Proceedings of the 8th Conference on Heritage Materials Science*, pages 114–121. Heritage Press, 2021. Sample entry for rendering checks.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

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Yumi Tanaka and Kenji Sato. Ink density saturation curves for traditional duan inkstones. *Journal of East Asian Material Culture*, 12(3):33–52, 2019. doi: 10.0000/jeamc.2019.12.33.